

SatQuery (SIH26167): Implementation Risks, Architecture Strategy, & Development Roadmap

Transitioning from a generic conversational chatbot wrapper to an enterprise-grade multi-modal Remote Sensing AI system. This blueprint maps core problem statement requirements, addresses 20 foundational technical risks, defines tool routing boundaries, and sets a sequential execution path.

10

Core PS Capabilities

20

Mapped Risks &
Solutions

3

Core Pipelines

8

Sequential Phases

SECTION 1

Context & Overview

With the official SIH26167 requirements established, the technical challenge transitions from building a generic conversational interface to engineering a robust, multi-modal Remote Sensing AI system.

CORE PROBLEM STATEMENT (PS) CAPABILITY REQUIREMENTS

1

Single-image VQA

Visual question answering over optical/SAR data.

2

Grounding / Captioning

Bounding region highlights or image summary.

3

Bi-temporal Change

Multi-date comparative change analysis.

4

Optical-SAR Analysis

Cross-modal paired fusion & detection.

5

Remote-sensing Adaptation

6

Input Inspection

Tailored model pathways for geospatial domain.

GeoTIFF CRS, bands, and dimension checks.

7

Specialist Selection

Strict registry tool routing engine.

8

Output Integration

Unified schema across text, masks & boxes.

9

Confidence Estimation

Measurable spatial & model score metrics.

10

Visual Evidence & Trace

Observable execution logs and grounded boxes.

Core Insight: The primary risk is not the chatbot wrapper itself, but ensuring that all these underlying specialized components operate reliably together.

SECTION 2

Comprehensive Risk Matrix

Detailed breakdown of all 20 implementation problems, structural difficulty, and concrete practical solutions.

Risk Category Breakdown

Distribution of technical challenges across system domains.



Accuracy & AI Data & Format Agent Routing & Pipeline

#	PROBLEM	WHY IT IS DIFFICULT	PRACTICAL SOLUTION
1	Remote-sensing VLM accuracy	Generic VLMs can misunderstand RS imagery; even GeoChat notes that general-domain VLMs can produce inaccurate or fabricated answers on RS queries.	Start from a remote-sensing model and adapt/evaluate it on the required datasets rather than relying on a general VLM.
2	SAR understanding	SAR is fundamentally different from ordinary RGB/optical imagery. Recent work still reports that optical-domain knowledge does not transfer cleanly to SAR.	Give SAR its own preprocessing/model pathway and test SAR questions separately.
3	Huge satellite images	A satellite GeoTIFF can be much larger than the input resolution expected by a VLM. Naively resizing the whole image can destroy small-object information.	Use tiling/cropping plus a coarse-to-fine strategy: locate relevant regions first, then analyze those regions at higher resolution.
4	GeoTIFF complexity	You cannot treat every TIFF as an ordinary RGB picture. Band count, datatype, CRS, resolution and metadata can vary.	Build an input inspection layer before inference. Detect modality, dimensions, bands, CRS and supported format.
5	Optical + SAR alignment	The PS specifically requires co-registered pairs. Even small spatial misalignment can damage pixel/region-level fusion.	Validate dimensions, spatial metadata and alignment before fusion; reject or flag incompatible pairs.
6	Bi-temporal change detection	Differences can come from genuine change, registration errors, illumination, seasonality or sensor differences.	Separate change detection from change interpretation and add registration/quality checks.
7	VQA hallucination	A language model can generate a convincing answer even when the visual evidence is weak.	Require evidence-producing specialist outputs and make the final response depend on them.
8	Grounding errors	The answer may be correct but the highlighted region may be wrong, or vice versa.	Evaluate text correctness and spatial correctness separately.
9	Agent chooses wrong model	A language model may misunderstand whether a query requires VQA, grounding, temporal analysis or cross-modal analysis.	Don't let the LLM freely invent tools. Give it a fixed capability registry containing supported inputs, tasks and outputs.

#	PROBLEM	WHY IT IS DIFFICULT	PRACTICAL SOLUTION
10	Too many models	Combining many research models can create dependency, GPU-memory and integration problems.	Start with the minimum model set required by the PS. Add another model only when experiments show a measurable benefit.
11	Different model output formats	One model may return text, another boxes, another masks, another scores.	Define one internal output schema for all specialist models.
12	Confidence estimation	"The LLM sounds confident" is not a valid confidence measure.	Build confidence from measurable evidence: model score, agreement, spatial validity, input quality and task-specific checks.
13	Data mismatch	Public datasets such as VRSBench/RSVQA do not perfectly reproduce the organizer's Cartosat/RISAT evaluation environment.	Keep a strict separation between public development/benchmark data and the expected ISRO/SAC evaluation domain.
14	Training cost	Some remote-sensing VLMs are expensive to fine-tune. GeoChat's published training instructions, for example, describe multi-GPU training.	Prefer parameter-efficient adaptation, inference-time composition, or existing checkpoints where licensing and hardware permit.
15	Latency	Running several large models for every question could make the demo painfully slow.	Use query-dependent routing. A simple VQA question should not trigger the temporal and SAR pipelines.
16	Model licensing	A research model may have code available but different restrictions for weights/data.	Audit the license of every model, dataset and dependency before committing to it.
17	Benchmark leakage	Accidentally tuning against test examples can make results look better than they really are.	Freeze prescribed test sets and use separate development/stress sets.
18	Agent trace	The PS explicitly expects an observable execution summary.	Log task selected, models/tools selected, relevant parameters and outputs in a clean machine-readable structure.

#	PROBLEM	WHY IT IS DIFFICULT	PRACTICAL SOLUTION
19	Hardware limitations	Running multiple VLMs, SAR models and image-processing pipelines simultaneously may exceed your available GPU/VRAM.	Design the system so models run sequentially and unload when unnecessary; benchmark memory early.
20	Overengineering	Your team could spend weeks building fancy features that aren't evaluated.	Treat the PS's mandatory capabilities as the product boundary.

SECTION 3

Top Priority Areas: 3 Problems to Test First

Before building UI components, validate these three core operational pillars.

A

Image Understanding Baseline

Verify fundamental interpretation reliability across modalities before proceeding:

- **Optical Imagery:** Is the baseline accuracy reliable enough?
- **SAR Imagery:** Is SAR signal interpretation accurate and robust?
- **Optical + SAR Fusion:** Can the system interpret both modalities together reliably?

Critical Note: Recent research shows SAR requires specific adaptation; optical understanding does not transfer cleanly to SAR.

B

Three Required Workflows

Establish and independently test three separate backend processing pipelines:

1. Single Image:
Single Image → VQA → Grounding / Captioning

2. Bi-temporal Pair:
Bi-temporal Pair → Change Detection → Change Explanation / VQA

3. Optical + SAR Pair:
Optical + SAR → Cross-Modal Analysis

Treat these as three independently testable pipelines. Put the agent on top only after each works.

C

Tool Routing via Agent Registry

Restrict tool selection using a strict contract-driven tool registry to prevent LLM hallucination or invented tools.

Tool: RemoteSensingVQA

Tool: Grounding

Tool: ChangeAnalyzer

Tool: OpticalSARAnalyzer

Safety Rule: Agent can only select tools whose declared input/output contracts match the user query.

TOOL REGISTRY SPECIFICATION & QUERY ROUTING EXAMPLES

Registry Contract Specs:

- RemoteSensingVQA: Input [One Optical/SAR] → Output [Textual answer]
- Grounding: Input [One Image + Text query] → Output [Bounding regions]
- ChangeAnalyzer: Input [Registered Pair] → Output [Change map + features]
- OpticalSARAnalyzer: Input [Optical + SAR Pair] → Output [Joint findings]

Query Routing Examples:

- "What is visible in this image?" → Single-image VQA
- "Highlight the water body." → Grounding
- "What changed between these two dates?" → Temporal Pipeline
- "Compare optical and SAR reveal." → Cross-modal Pipeline

SECTION 4 & 5

System Architecture & Model Benchmarking

Structured verification and execution backend flow replacing unconstrained LLM decision making.



CANDIDATE MODEL STRATEGY

Do not assume pre-existing models work seamless out-of-the-box. Evaluate these candidate models on target imagery early:

GeoChat: Conversational remote-sensing VQA, captioning, and grounding.

EarthGPT: Multi-sensor remote-sensing comprehension (optical + SAR).

Tool-augmented RS-VLMs: Supporting dual optical/SAR input frameworks.

EARLY BENCHMARK ACTION ITEM

Benchmark candidate models on target sample imagery early to evaluate actual performance versus paper claims before finalizing architecture choices.

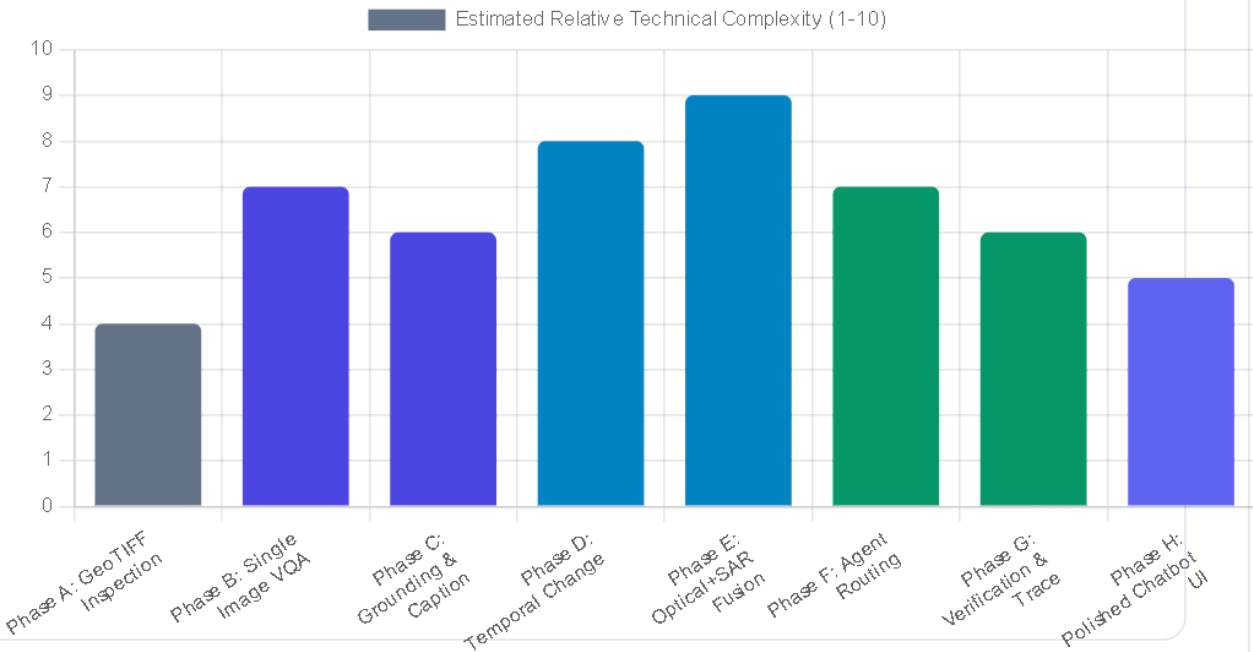
Key takeaway: Otherwise team could spend most of the hackathon integrating a model that looks impressive in paper but performs poorly on actual competition inputs.

SECTION 6

Phase-by-Phase Development Order

Chronological execution plan prioritizing backend capability before frontend interface building.

Phase Complexity & Sequence Visualizer



GeoTIFF Inspection

PHASE A

Can we load and correctly inspect GeoTIFF/TIFF metadata, bands, CRS, and dimensions?

Single-Image VQA

PHASE B

Can we run reliable single-image visual question answering on optical and SAR data?

Grounding & Caption

PHASE C

Can we generate bounding region highlights and descriptive captions accurately?

Temporal Change

PHASE D

Can we detect, isolate, and describe temporal change across multi-date image pairs?

Optical + SAR Fusion

PHASE E

Agent Tool Routing

PHASE F

Can we meaningfully analyze co-registered optical and SAR imagery together?

Can an agent controller correctly route queries matching tool input/output contracts?

Verification & Trace

PHASE G

Can we verify answers, estimate confidence scores, and generate execution traces?

Polished Chatbot UI

PHASE H

Build the final interactive chatbot UI wrapping all underlying backend pipelines.

STRATEGIC SUMMARY

The Problem Statement provides a clear feature boundary. The core technical objective is ensuring the underlying specialized remote-sensing pipelines execute reliably and integrate effectively on domain-specific satellite data.